

RESEARCH ARTICLE | FEBRUARY 12 2025

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Physics of Fluids 37, 023358 (2025)

<https://doi.org/10.1063/5.0252502>



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Cite as: Phys. Fluids **37**, 023358 (2025); doi: [10.1063/5.0252502](https://doi.org/10.1063/5.0252502)
Submitted: 10 December 2024 · Accepted: 16 January 2025 ·
Published Online: 12 February 2025



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ABSTRACT

Impact responses of granular materials remain poorly understood, posing significant challenges to extraterrestrial exploration activities such as landing, sampling, drilling, and construction. We studied the crater morphology formed in the granular media and their granular behavior on low-velocity impact cratering by integrating three-dimensional surface scanning, machine learning-based classification, and x-ray computed tomography (X-CT) imaging. Our laboratory experiments focused on lunar (LHS-1 and LMS-1) and Martian regolith simulants (MGS-1) and terrestrial fine silica sand (T-8) as studied granular materials. The profiles of formed craters were analyzed under various experimental conditions, defined by different spherical projectile diameters and fall heights, accounting for the low kinetic energy at the granular surface contact of 0.1–0.5 mJ. In machine learning, we applied logistic regression models to classify target granular materials based on crater profile features, revealing primary morphological differences influenced by the given granular properties. The particle distribution from X-CT imaging revealed notable differences in the granular behavior of regolith simulants due to impact compared with T-8. These results underscored significant differences between the granular properties of terrestrial fine sand and regolith simulants. Additionally, our X-CT data emphasized that the high cohesion of LMS-1 significantly enhanced its resistance to impact, resulting in the porosity decline beneath the crater bottom of 0.20 and 0.03 on average in the LHS-1 and LMS-1 media, respectively, when the kinetic energy was 0.3 mJ. Our findings highlight the distinctive granular properties of regolith particles, advancing our understanding of their granular responses to impact.

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I. INTRODUCTION

The impact behavior of granular materials has been extensively studied to gain crucial insights into their response to external forces.^{1–5} Impact cratering experiments, such as the launch of projectiles into target granular materials, provide valuable data on crater characteristics, including depth, diameter, and morphology. These crater features

are closely related to the properties of the given granular material, significantly influencing the granular physics and resulting behavior. Cook and Mortensen³ worked projectile impact experiments in dry sands, revealing the dependence of crater size on projectile and target granular material properties. The subsequent studies on low-velocity impacts revealed that the speed of ejected projectile scales with the

projectile-to-target grain size ratio and the impact speed.^{6,7} Additionally, the crater depth follows a power-law scaling with projectile sphere density, diameter, and drop height when the sphere is not entirely buried.^{8–11} These findings have advanced our understanding of granular behavior under impact conditions. However, little is understood about how the physical and mechanical properties of granular materials relate to crater morphology, such as curvature and roughness.

The physical and geomechanical properties of lunar and Martian surfaces remain a fundamental scientific question. The lunar and Martian surfaces are blanketed by a layer of loose granular deposits “regolith,” ranging from a few to 10s meters thick on average on the lunar surfaces. This regolith layer on the lunar surface is continually bombarded by micrometeorites, leading to fragmentation, partial melting, and forming irregular clusters known as agglutinates.¹² Martian regolith undergoes similar processes, with additional complex geological influences.¹³ These processes cause regolith particles to exhibit specific granular properties distinct from Earth’s sand particles. Numerous missions have explored and sampled these lunar and Martian surfaces. The missions have revealed the inherent physical and geomechanical properties posing significant challenges data interpretation, landing and drilling operations, and constructing infrastructure, highlighting the necessity of a thorough understanding of the granular properties of regolith.¹⁴ Due to difficulties in sufficient volume of actual regolith samples,^{15–17} regolith simulants have been developed to replicate the physical, mechanical, and chemical properties of actual regolith. Studies have validated the suitability of regolith simulants, like LHS-1, LMS-1, and MGS-1 for lunar highlands, lunar mare, and Martian basaltic regolith, respectively.^{15–17} These high-fidelity simulants enable controlled laboratory experiments to study various granular properties of a given regolith. However, very few publications have studied changes in the porous microstructure and porosity of granular regolith simulants, along with the impact cratering on the regolith surface.

This study focuses on the crater morphology formed by low-velocity projectile impacts on granular regolith simulant media. We integrate experimental observations, three-dimensional (3D) surface scanning, machine learning, and x-ray computed tomography (X-CT) measurements. First, 3D surface scanning is performed to obtain high-resolution surface profiles of craters generated by projectile impacts. Our studied craters are simple-shaped craters representing a “bowl shape” because of the low-velocity impact cratering. Second, machine learning is conducted to examine whether we can classify different granular materials (regolith simulants and terrestrial sand) based on obtained crater profiles. This approach enables the systematic and objective characterization of granular media properties, as documented in the morphology of the resulting craters. Third, X-CT measurements are done to image the difference between granular-pore microstructures before and after projectile impacts. Previous studies have predominantly focused on surface-level metrics such as crater depth, radius, shape, and ejecta patterns. The X-CT imaging allows us to examine the subsurface responses to impact cratering, which cannot be studied by 3D surface scanning alone, offering a more comprehensive understanding of granular material behavior beneath the crater surface. By integrating datasets of crater profiles and X-CT, this study aims to establish a connection between surface characteristics in crater morphology and geomechanical properties of the given granular material, such as porosity and cohesion. Furthermore, classifying target

granular materials based on shallow crater morphologies and pore-particle microstructure analysis with X-CT imaging provides critical insights into the geomechanical responses of the granular materials to impact. This study will also provide insights for investigating granular material properties from crater profiles alone on the lunar and Martian surface, helping to develop effective exploration, excavation, and resource utilization strategies for future extraterrestrial missions.

II. MATERIAL AND METHODOLOGY

A. Studied granular materials

This paper studied four granular materials with different physical properties (Table I). The studied granular materials were terrestrial silica sand Tohoku Keisa No. 8 (T-8) and three different high-fidelity regolith simulants developed by Exolith Lab. This type of silica sand has well-documented and consistent geomechanical properties comparable to those of lunar and Martian regolith simulants (Table I), allowing for comparison with the behavior of extraterrestrial regolith simulants under controlled experimental conditions.^{6,7,18,19} The high-fidelity regolith simulants were the lunar highlands regolith simulant LHS-1, lunar mare regolith simulant LMS-1, and Martian global regolith simulant MGS-1.^{15,16} Various studies have revealed that the studied regolith simulants well replicate the mineral, chemical, and geomechanical properties of the actual lunar and Martian regolith particles.^{17,20}

We filled the given studied granular material in acrylic containers for the impact crater experiments. Following the method by Ishii *et al.*,¹¹ we prepared the granular media for the respective experiment by carefully filling the given granular material into the container. The mean porosity $\bar{\phi}$ of the given granular media filled in the container averaged over the height of the container was estimated by

phi_bar = (V_c - M_g / rho_g) / V_c, (1)

where V_c is the container volume, M_g is the mass of the given granular media within the container, and ρ_g is the mean particle density of the given granular material.

B. Impact cratering experiments

The goal here is to examine whether we can classify the granular media and investigate granular behavior upon impact using the most

TABLE I. Physical properties of studied granular materials.

Target material	Mean particle density ρ_g (g/cm ³)	Mean particle diameter (μ m)	Cohesion (Pa)
LHS-1	3.23	90 ^a	311 ^b
LMS-1	3.34	91 ^c	393 ^b
MGS-1	3.40	90 ^d	307 ^e
T-8	2.78 ^{f,g}	120 ^{f,g}	0 ^f

^aExolith Lab.²¹
^bMillwater *et al.*²²
^cExolith Lab.²³
^dExolith Lab.²⁴
^eDotson *et al.*²⁵
^fOzaki *et al.*¹⁹
^gTabuchi *et al.*²⁶

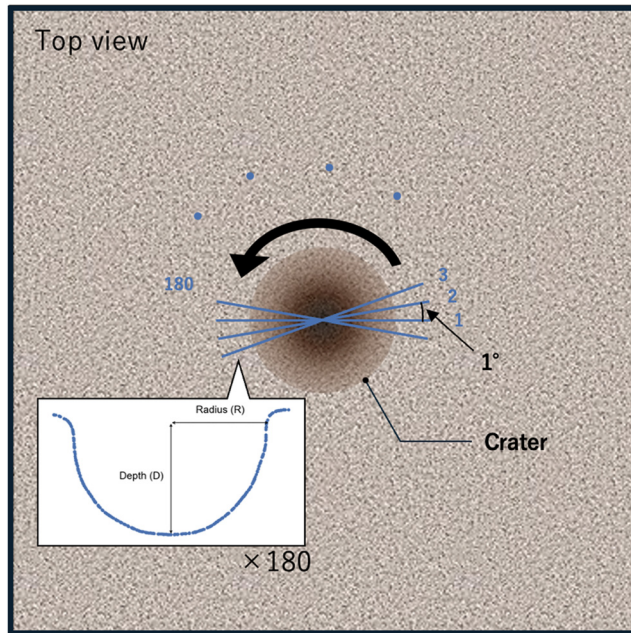


FIG. 1. Schematic diagram illustrating how we obtained 180 profiles from the given crater.

simple and generic lab experiments possible. In a series of impact cratering experiments, we, thus, used spherical alumina projectile (3.6 g/cm^3) HD-10 and HD-8 (Nikkato Corporation, Osaka, Japan), with a diameter of 10 and 8 mm, respectively. Following Ishii *et al.*,¹¹ we released the spherical alumina projectile vertically without rotation to generate craters upon impact under the earth's gravity in terrestrial ambient conditions at $\sim 25^\circ\text{C}$. We dropped the spherical projectile immediately after preparing the given target granular layer in the container using a height-adjustable clamp. The studied drop heights were 10.0 and 15.0 mm. The resulting impact velocities on the surface of granular media were 0.44 m/s (8 mm diameter projectile and 10.0 mm drop height), 0.44 m/s (10 mm diameter projectile and 10.0 mm drop height), 0.54 m/s (8 mm diameter projectile and 15.0 mm drop height), and 0.54 m/s (10 mm diameter projectile and 15.0 mm drop height). The resulting kinetic energy at the surface contact ranged from 0.1 mJ to 0.3 mJ. The ratio of the container size to the studied spherical projectile size is high enough to ignore the possible influence on the penetration depth.²⁷ This study conducted eight impacts at the given condition to assess their uncertainties in generated crater morphology, resulting in a total of 128 impact cratering experiments.

C. Cross-sectional crater profiles

We used a CR-Scan Lizard 3D Scanner (Shenzhen Crealty 3D Technology Co., Ltd., China) to acquire high-resolution 3D digital surface data of the granular media after removing the spherical projectile meticulously with a vacuum pickup tool (Nihon Pisco Co., Ltd.,

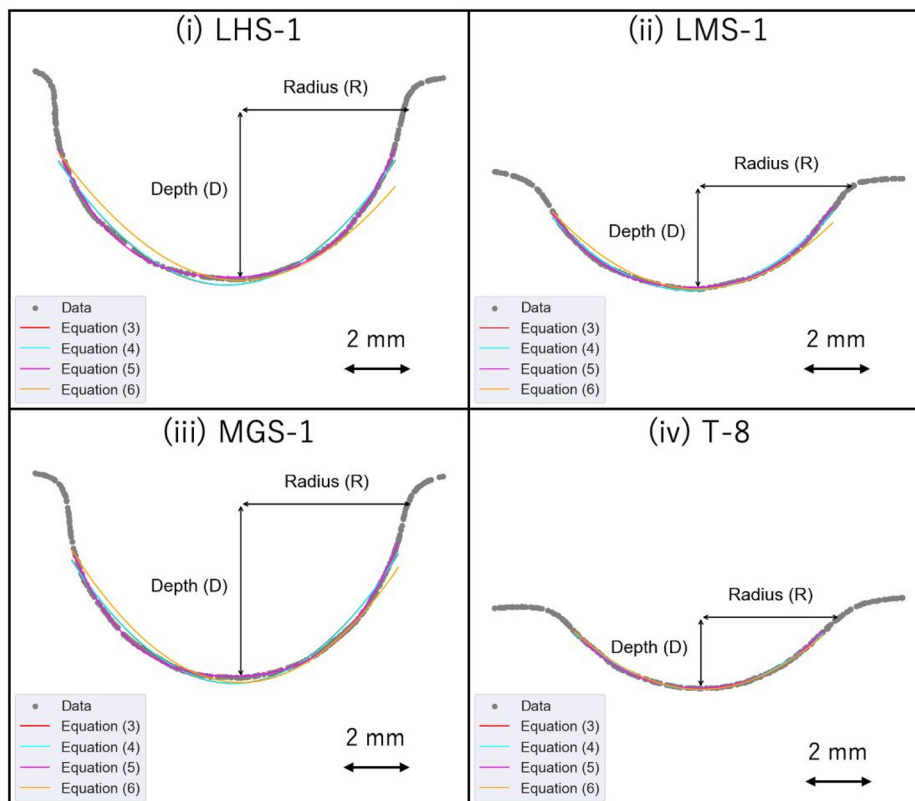


FIG. 2. Crater profiles formed by the spherical projectile with a diameter of 10 mm dropped from 10 mm above the surface of (i) LHS-1, (ii) LMS-1, (iii) MGS-1, and (iv) T-8.

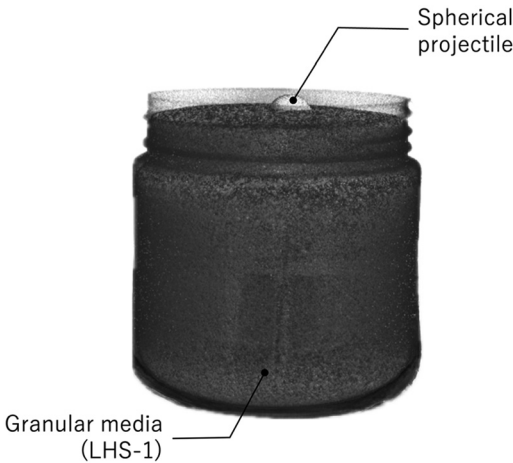


FIG. 3. Studied object visualized by X-CT imaging.

Nagano, Japan). The accuracy and resolution of the obtained 3D digital surface data were 50 and 100 μm , respectively. We then converted the acquired 3D digital surface data into point cloud data using the open-source software CloudCompare. To analyze the morphology of the generated crater, we obtained a total of 23 040 cross-sectional crater profile datasets by generating 180 cross-sectional profiles from the given crater at a 1-degree increment around its deepest point (Fig. 1).

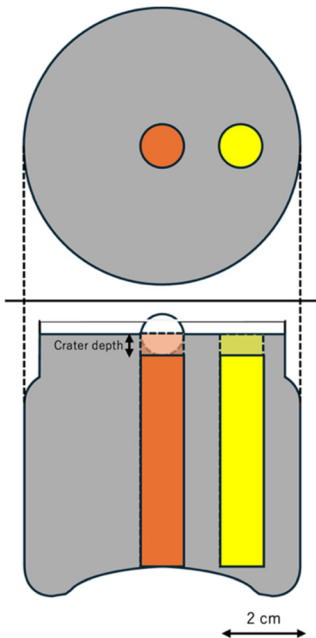


FIG. 4. Two studied cylindrical regions in the X-CT image data. Locations below the sphere (Center; orange) and the midpoint between the lateral end of sphere and the farthest wall of the container (Side; yellow). The translucent area in the uppermost granular media surface is included in the studied area for only the target material before impact.

TABLE II. The model performance under conditions (a)–(d).

Condition	a	b	c	d
Training score	0.841	0.797	0.812	0.782
Validation score	0.836	0.804	0.798	0.789
Logistic regression test accuracy	0.643	0.619	0.522	0.292

TABLE III. LR confusion matrix for the classification under condition (a).

	LHS-1	LMS-1	MGS-1	T-8
LHS-1	2805	0	75	0
LMS-1	418	1203	1259	0
MGS-1	2058	11	811	0
T-8	0	236	55	2589

TABLE IV. LR confusion matrix for the classification under condition (b).

	LHS-1	LMS-1	MGS-1	T-8
LHS-1	1102	771	1007	0
LMS-1	0	2474	0	406
MGS-1	167	1832	881	0
T-8	0	206	0	2674

TABLE V. LR confusion matrix for the classification under condition (c).

	LHS-1	LMS-1	MGS-1	T-8
LHS-1	30	152	2698	0
LMS-1	0	2073	721	86
MGS-1	1	256	2623	0
T-8	0	1594	0	1286

TABLE VI. LR confusion matrix for the classification under condition (d).

	LHS-1	LMS-1	MGS-1	T-8
LHS-1	350	1610	159	761
LMS-1	0	147	0	2733
MGS-1	378	1097	40	1365
T-8	53	0	0	2827

To characterize the crater morphology, we calculated the aspect ratio (AR) of the given crater defined by

$$AR = 2R/D, \tag{2}$$

where R and D are the radius and depth of the given crater (Fig. 1). The crater depth (D) was determined relative to the median level of the surface of the granular bed surrounding the crater, with the deepest

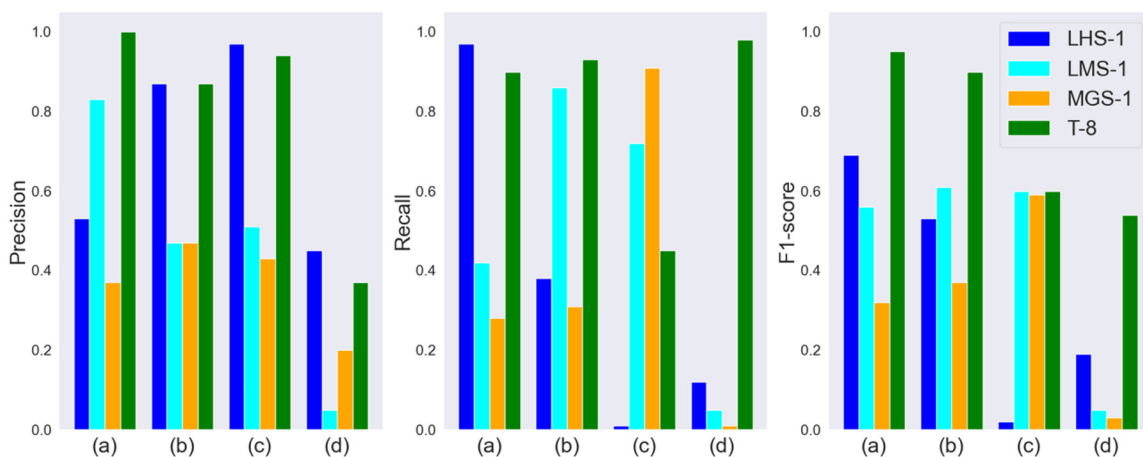


FIG. 5. Precision, recall, and F1-score of each classified target material under conditions (a)–(d).

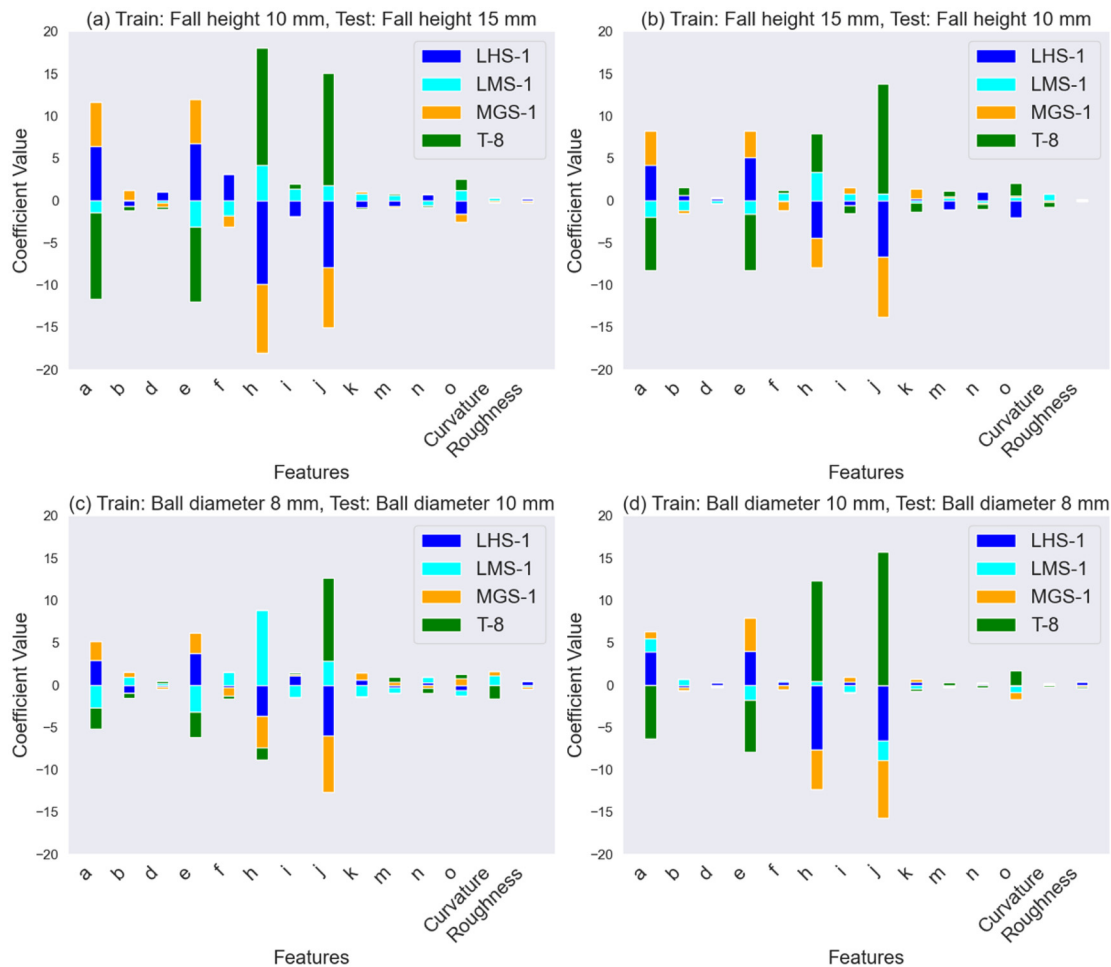


FIG. 6. Coefficients of each feature to classify the target materials. (a) Each coefficient for the model trained on craters formed by a 10 mm drop. (b) Each coefficient of the model trained on craters formed by a 15 mm drop. (c) Each coefficient of the model trained on craters formed by an 8 mm sphere. (d) Each coefficient of the model trained on craters formed by a 10 mm sphere.

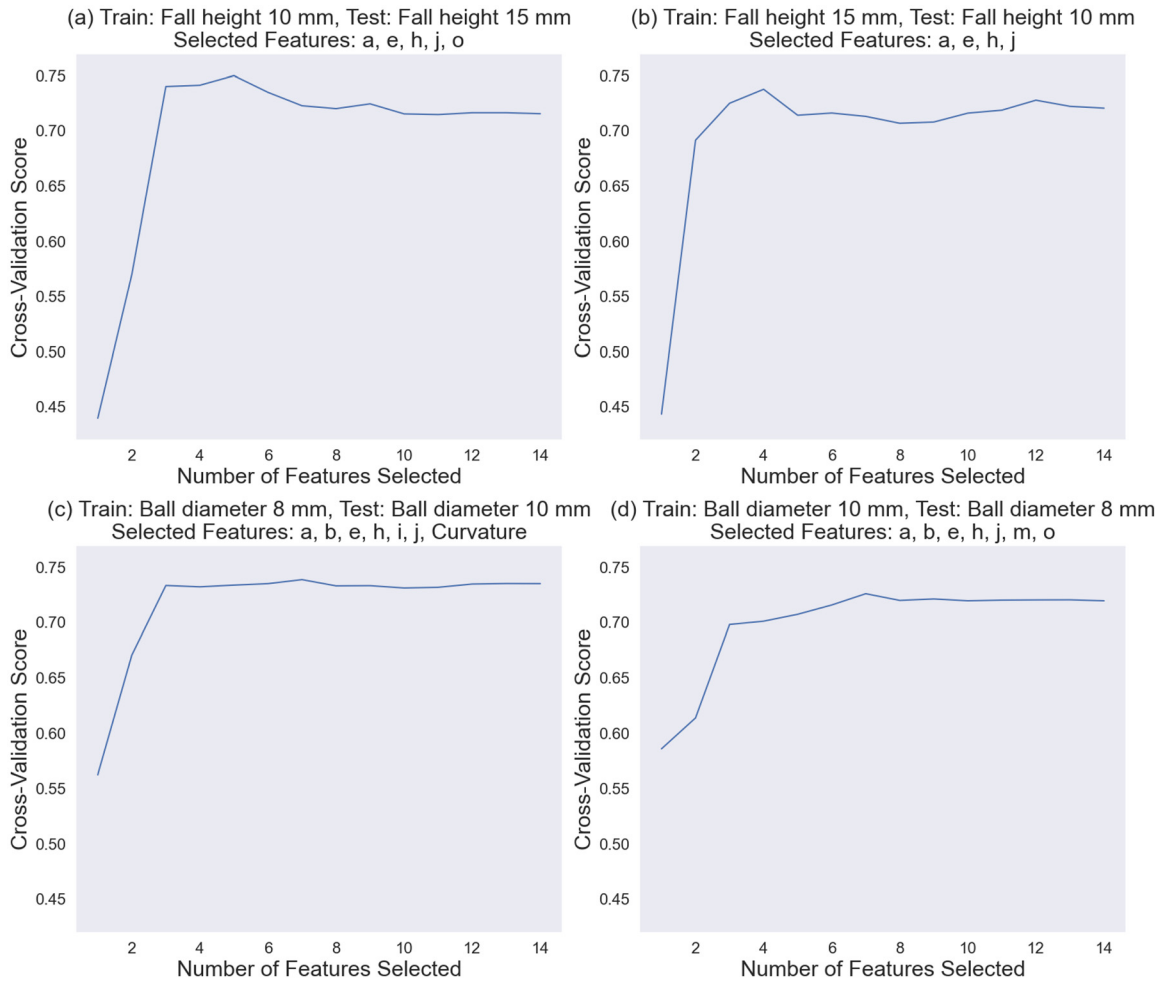


FIG. 7. RFECV for optimizing machine learning models. (a) Five features (a , e , h , j , and o) optimize the model trained on craters formed by a 10 mm drop. (b) Four features (a , e , h , and j) optimize the model trained on craters formed by a 15 mm drop. (c) Seven features (a , b , e , h , i , j , and Curvature) optimize the model trained on craters formed by an 8 mm sphere. (d) Seven features (a , b , e , h , j , m , and o) optimize the model trained on craters formed by a 10 mm sphere.

point marking the crater bottom. The crater radius (R) was defined as the radius of the circle formed by the cross section of the sphere at this referenced surface (Fig. 1). When the crater depth (D) exceeded the sphere radius, the sphere radius was used as the reference for the crater radius. We studied only the condition satisfying $D < 2R$ for cross-sectional crater profiles that could preserve the resulting crater morphology without any granular collapses.

D. Machine learning with crater profiles

Curve fitting has been applied to a range of crater profiles, from relatively simple parabolic shapes to more complex geometries.^{28–30} Because we study the low-velocity impact, our craters are simple craters with no central peak representing a “bowl shape.”³¹ Thus, we can fit the crater profiles away from the rim of crater using various curve fitting models. Among the commonly used curve fitting models for crater profiles, we used second-, third-, and fourth-order polynomial and hyperbola functions to obtain coefficients for each approximation

$$ax^2 + bx + c, \quad (3)$$

$$dx^3 + ex^2 + fx + g, \quad (4)$$

$$hx^4 + ix^3 + jx^2 + kx + l, \quad (5)$$

$$m + \sqrt{n^2 + o^2x^2}, \quad (6)$$

where x is the lateral distance from the deepest point of the given crater, and a , b , c , d , e , f , g , h , i , j , k , l , m , n , and o are coefficients of the given fitting equation. We performed curve fitting to the partial crater profile that fell within the lowest point and the 80% depth level of the crater depth (Fig. 2). We used each coefficient except for c , g , and l as explanatory variables for machine learning to avoid the influence of the crater depth and focus on crater shape. Additionally, we examined the average roughness and curvature of the points comprising a crater’s cross section defined by the positional correlation between adjacent points.

We then conducted two training and testing sessions using a machine learning model. In our machine learning model, we used

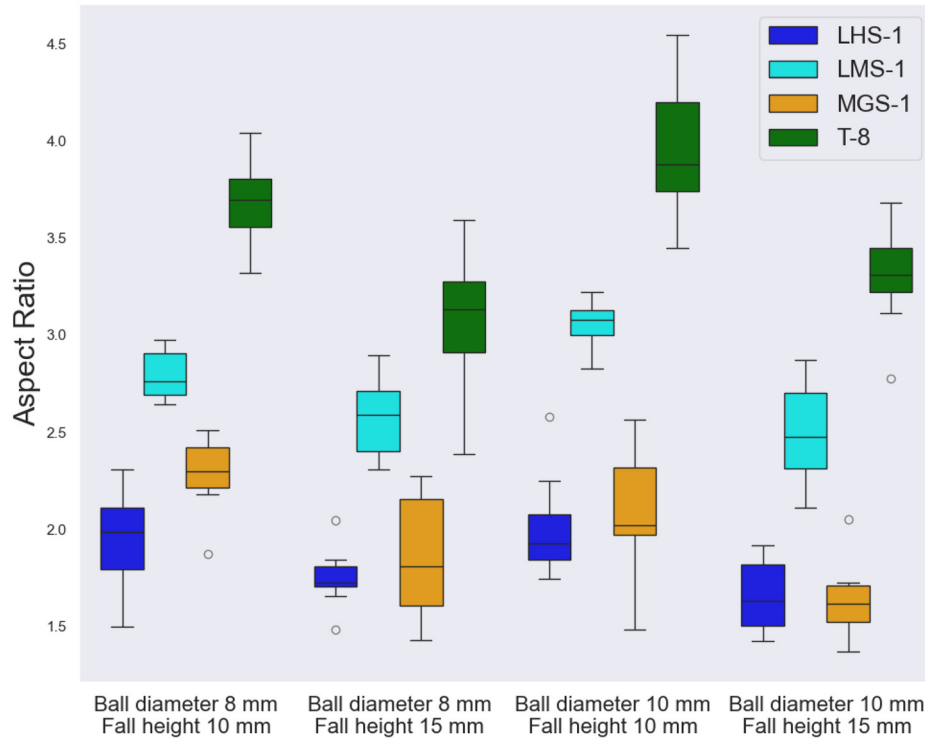


FIG. 8. Aspect ratio boxplot by ball diameter, fall height, and target material. The aspect ratio (AR) is calculated by Eq. (2).

multiclass Logistic Regression (LR) to examine whether we can classify and predict the type of granular media (i.e., LHS-1, LMS-1, MGS-1, and T-8) from the crater morphology represented by a series of coefficients in Eqs. (3)–(6), roughness, and curvature. The multiclass LR model (with K classes) in the probability of an observation belonging to class k (p_k) is expressed by

$$z_k = \beta_{k,0} + \sum_{i=1}^n \beta_{k,i} X_i, \quad (7)$$

$$p_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}, \quad \sum_{k=1}^K p_k = 1, \quad (8)$$

where $\beta_{k,0}$ is the intercept term for class k ($k = 1, 2, \dots, K$) of the model (constant), and $\beta_{k,i}$ is the coefficient for each feature X_i for class k derived from crater features. LR is widely recognized for its high interpretability, as it allows for calculating the relative importance of features through their coefficients. To leverage this interpretability, we investigated the influence of coefficients β_i for each feature and conducted Recursive Feature Elimination with Cross-Validation (RFECV). This approach enabled to quantify each feature's contribution to the model performance and systematically identify the important features, thereby enhancing the robustness and interpretability of our results. First, we trained the models with the explanatory variables of craters created by spheres dropped from 10 and 15 mm above the surface and tested them on craters created by spheres dropped from 15 and 10 mm above the surface, respectively. Second, we trained the models with the explanatory variables of craters created by spheres with a diameter of 8 and 10 mm and tested them on craters created by spheres with a diameter of 10 and 8 mm, respectively. The machine

learning models then classified the target granular material as a response variable in both testing processes based on the explanatory variables derived from the crater characteristics.

E. Acquisition and analysis of x-ray computed tomography (X-CT) images

To better characterize the internal microstructure of the studied granular media, we performed X-CT imaging for selected granular media samples. We used a Geotek Rotating X-CT (RXCT) at STRIDE-C of National Taiwan University. The cylindrical plastic bottle with a diameter of 65 mm and a height of 65 mm was filled with studied granular media and fixed in a polyvinyl chloride pipe to stabilize the granular media sample while scanning in RXCT. The high-resolution X-CT imaging of the granular media could be obtained before and after generating craters due to the ability of the RXCT system to perform scans without moving the granular material. The granular media after impact crater experiment were scanned without removing a dropped spherical projectile in RXCT (Fig. 3). The X-CT reconstruction was performed using a Geotek reconstruction software. The resulting resolution of all the acquired X-CT images was $57 \mu\text{m}$.

The acquired X-CT image data were analyzed using Dragonfly software (Comet Technologies Canada Inc., Montréal, Canada). In the software, both pre- and post-impact X-CT image sets of the granular materials were imported into the same workspace for accurate comparison purposes. Intensity values were calibrated to standardize the background and foreground intensities at 0 and 100, respectively, ensuring consistency across datasets. Although this calibration method was optimized, slight discrepancies might exist due to the automatic

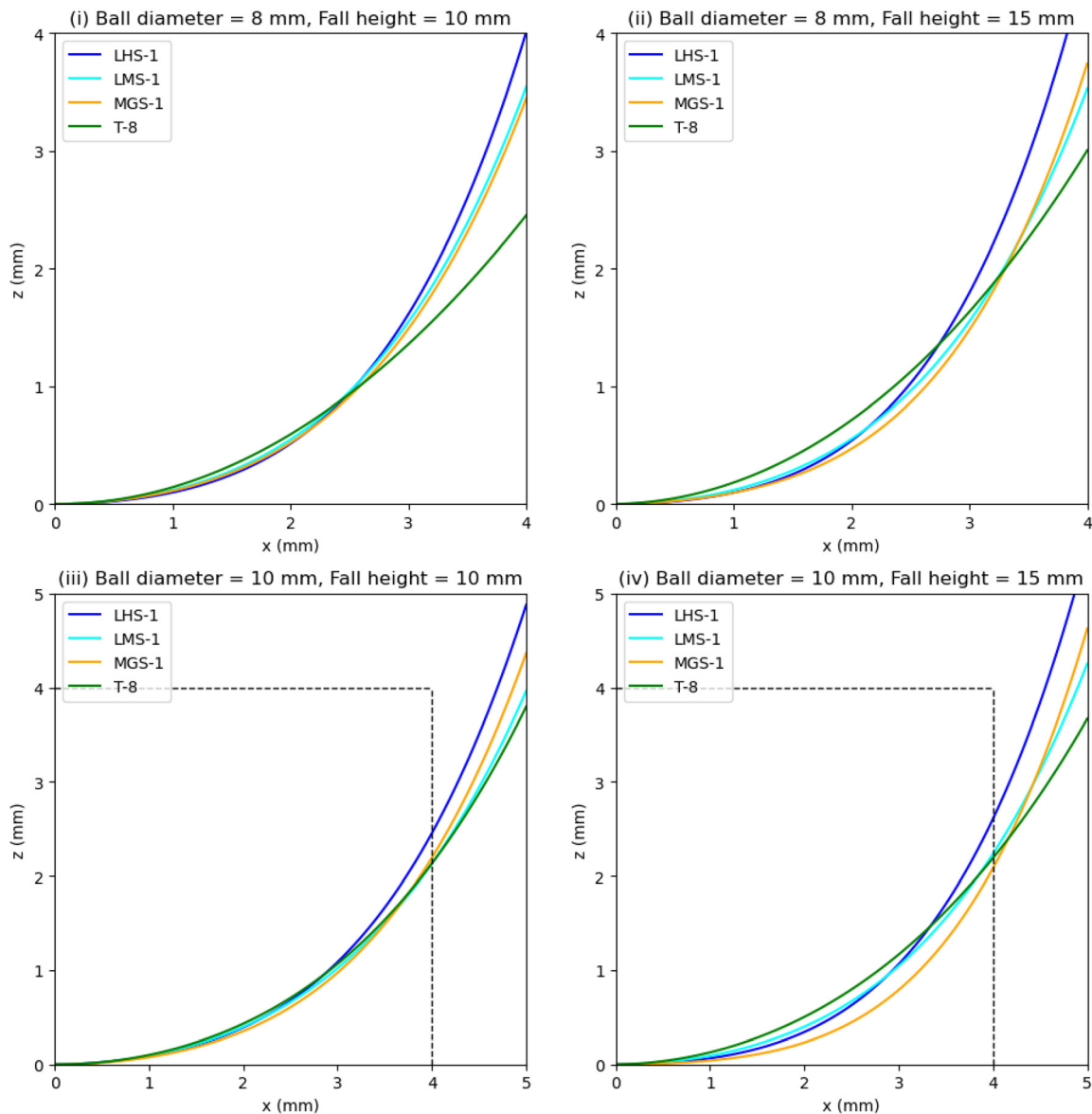


FIG. 9. Half of the crater profile derived from the average of the obtained coefficients h , i , j , and k using Eq. (5) under different experimental conditions. (i) Spherical projectile with a diameter of 8 mm dropped from 10 mm above the surface. (ii) Spherical projectile with a diameter of 8 mm dropped from 15 mm above the surface. (iii) Spherical projectile with a diameter of 10 mm dropped from 10 mm above the surface. (iv) Spherical projectile with a diameter of 10 mm dropped from 15 mm above the surface. The dot lines are references to compare across the craters formed by spheres with different diameters.

classification of the background (primarily air) and foreground (pipe, container, and granular media) when working in the software.

A hypothetical void model was established by determining the intensity representing voids within each granular material before impact, aligning its porosity with the mean porosity (ϕ) physically measured using Eq. (1). Furthermore, the particles in the pre- and post-impacted target layers were identified based on the intensity thresholds for the voids, enabling a visual comparison of particle distributions across different granular materials and between pre- and

post-impact states. In this study, we focused on the two regions of the cylinder column with a diameter of sphere (Fig. 4); locations beneath the impact sphere (“center”) and at the midpoint between the lateral end of the sphere and the container’s farthest wall (“side”). The intensity and porosity of the target material before impact were measured from its surface to a depth of 50 mm, and those after impact were measured from a position deeper than the level of crater depth. Eight data-sets, capturing the intensity and porosity values in the studied regions, were visualized to comprehensively analyze differences in studied

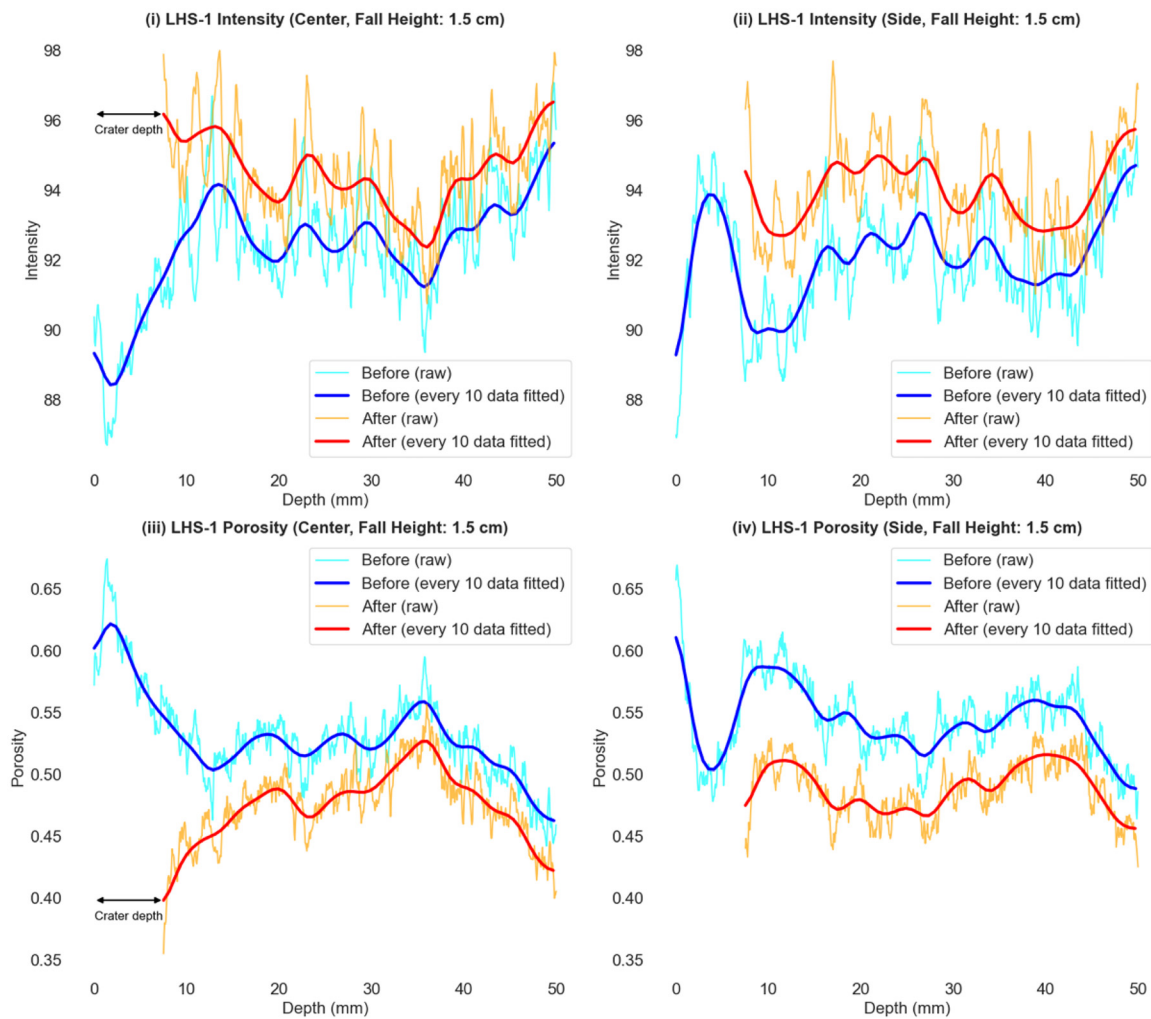


FIG. 10. Comparison of intensity profiles of LHS-1 between before and after impact (1.5 cm drop) in the studied area, (i) “center” and (ii) “side” (refer to Fig. 4). Comparison of hypothetical porosity profiles of LHS-1 between before and after impact (1.5 cm drop) in the studied area, (iii) “center” and (iv) “side” (refer to Fig. 4). The generated crater depth was 7.5 mm.

granular media (i.e., LHS-1, LMS-1, MGS-1, and T-8) and pre- and post-impacts.

III. RESULTS AND DISCUSSION

A. Classification of target granular materials with LR models

We used multi-class LR machine learning models to examine whether we can classify and predict the studied granular materials (LHS-1, LMS-1, MGS-1, and T-8) from the crater morphology represented by a series of coefficients in Eqs. (3)–(6), roughness, and curvature. The results of the LR models under four different conditions (a)–(d), defined with the features extracted from craters formed by spheres with diameters of 8 mm or 10 mm and fall height of 10 mm or 15 mm, are presented by model performance (training and validation scores and test accuracy), confusion matrix, and classification

metrics (precision, recall, and F1-score) under the given condition (Tables II–VI and Fig. 5). The four conditions (a)–(d) are:

- The model was trained using impact craters formed by spheres dropped from a height of 10 mm and tested on craters formed by drops from 15 mm height,
- The model was trained using impact craters formed by spheres dropped from a height of 15 mm and tested on craters formed by drops from 10 mm height,
- The model was trained on impact craters created by an 8 mm sphere and tested on craters created by a 10 mm sphere,
- The model was trained on impact craters created by a 10 mm sphere and tested on craters created by an 8 mm sphere.

Among the four studied conditions, condition (a) showed the best training score, validation score, and accuracy, with 0.841, 0.836, and 0.643, respectively (Table II). The confusion matrix in condition

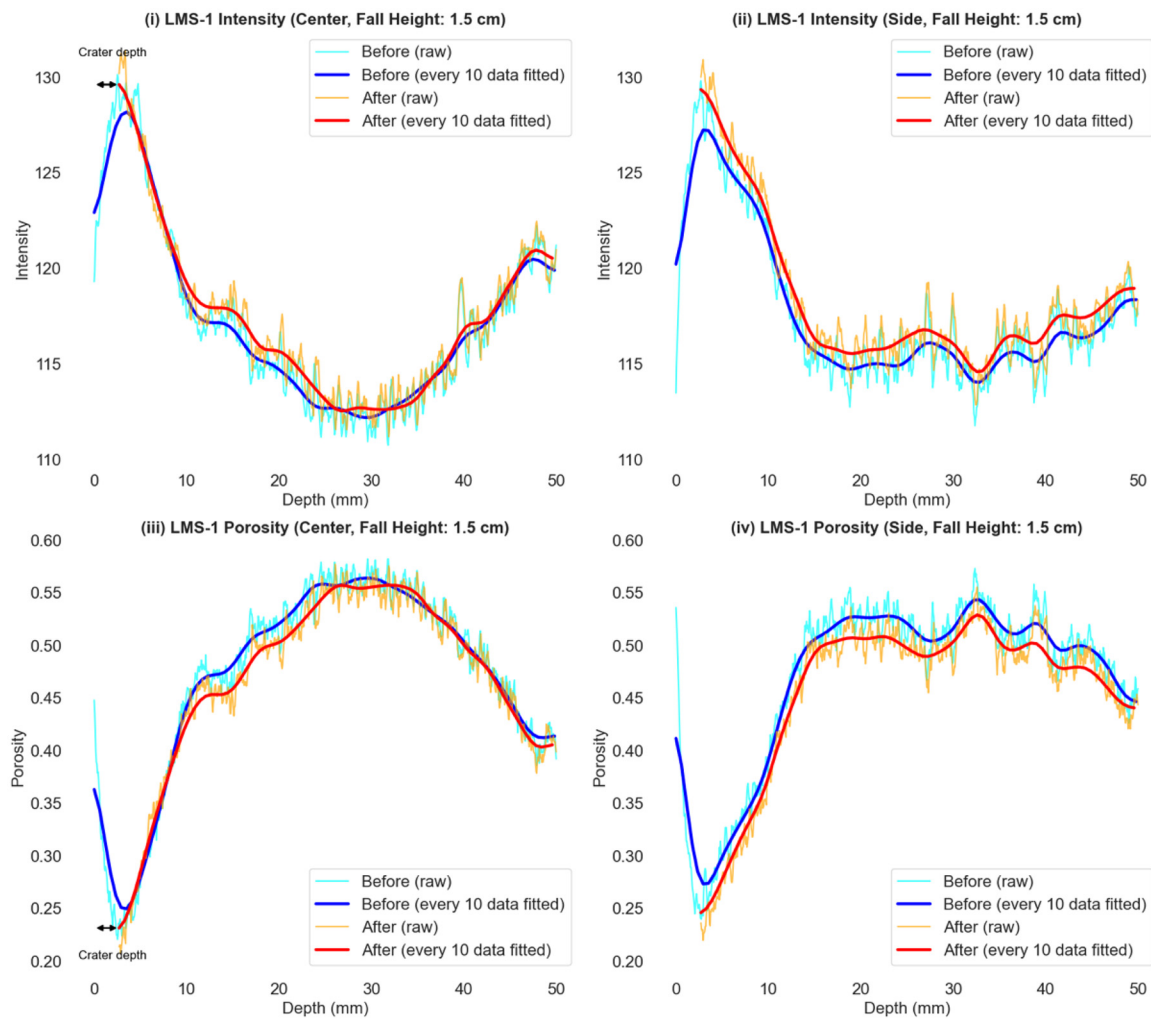


FIG. 11. Comparison of intensity profiles of LMS-1 between before and after impact (1.5 cm drop) in the studied area, (i) “center” and (ii) “side” (refer to Fig. 4). Comparison of hypothetical porosity profiles of LMS-1 between before and after impact (1.5 cm drop) in the studied area, (iii) “center” and (iv) “side” (refer to Fig. 4). The generated crater depth was 2.7 mm.

(a) revealed that our LR model well classified LHS-1 and T-8, but it was unsuccessful in LMS-1 and MGS-1 (Table III). The recall and F1 score were high in LHS-1 and T-8 and were low in MGS-1, whereas the precision was high in T-8 and LMS-1 (Fig. 5). Following condition (a), condition (b) showed a good accuracy of 0.619, with a training score and validation score of 0.797 and 0.804, respectively (Table II). The confusion matrix in condition (b) showed that our LR model well classified LMS-1 and T-8, but it was unsuccessful in LHS-1 and MGS-1 (Table IV). The recall and F1 score were high in T-8 and LMS-1 and were low in MGS-1, whereas the precision was high in LHS-1 and T-8 (Fig. 5). The results indicated that the successful classification of granular materials varied when drop heights of training and testing changed. Nevertheless, T-8 was well distinguished from the other granular media in both conditions (a) and (b), suggesting significant differences between the crater morphologies of T-8 and the other granular regolith media in the two conditions.

In contrast, conditions (c) and (d) showed lower accuracy, with 0.522 and 0.292, respectively (Table II). The confusion matrix in condition (c) showed that our LR model was good in classifying LMS-1 and MGS-1, while it was unsuccessful in LHS-1 and T-8 (Table V). The recall was high in LMS-1 and MGS-1 and was considerably low in LHS-1, whereas the precision was high in LHS-1 and T-8 and was low in MGS-1 (Fig. 5). The F1 score in LHS-1 was considerably low. The confusion matrix in condition (d) showed that our LR model could classify T-8, but it did not successfully classify LHS-1, LMS-1, and MGS-1 (Table VI). Although the recall in T-8 was high, the recall, precision, and F1 score in LHS-1, LMS-1, and MGS-1 were generally very low (Fig. 5). The results indicated that the successful classification of granular materials also varied when sphere diameters of training and testing changed. Moreover, our results suggested that the classification in different sphere diameters was more complex than in different drop heights, likely because the ratio of kinetic energy at the impact between

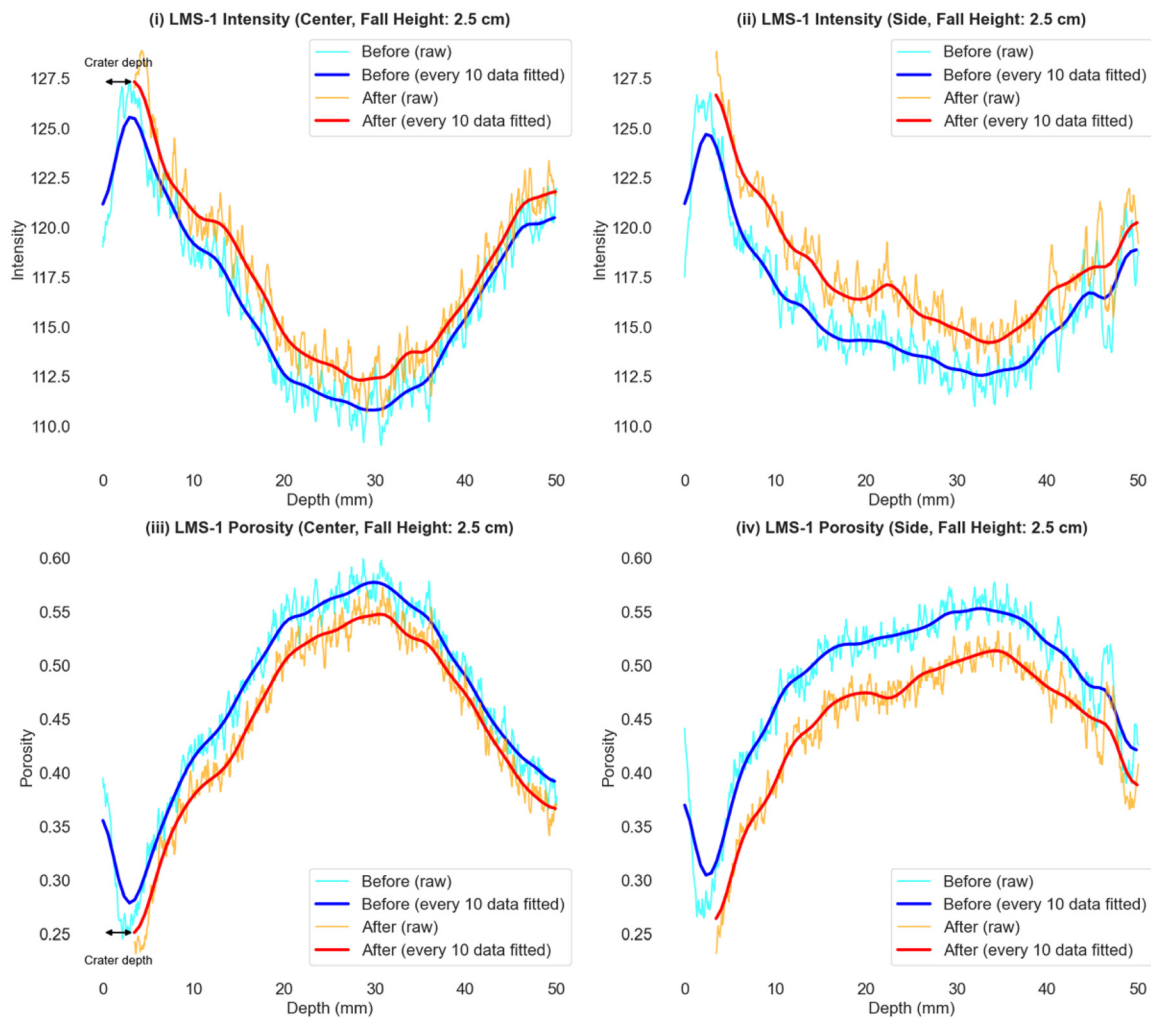


FIG. 12. Comparison of intensity profiles of LMS-1 between before and after impact (2.5 cm drop) in the studied area, (i) “center” and (ii) “side” (refer to Fig. 4). Comparison of hypothetical porosity profiles of LMS-1 between before and after impact (2.5 cm drop) in the studied area, (iii) “center” and (iv) “side” (refer to Fig. 4). The generated crater depth was 3.5 mm.

sphere diameters of 10 and 8 mm (1.95) is higher than that of fall heights of 15 and 10 mm (1.51). Nevertheless, T-8 was well distinguished from the other granular media in all the studied conditions (a)–(d), indicating significant differences between crater morphologies of T-8 and those of the other granular regolith media in all the conditions.

We also obtained values of the coefficients in Eqs. (3)–(6), mean curvature, and mean roughness representing the feature of crater morphology (Fig. 6) and RFECV to select the number of features and influential features to optimize the performance of our machine learning models (Fig. 7). The coefficients of the polynomial features, particularly the coefficients of second-order polynomials (a , e , and h) and fourth-order polynomial (j), consistently exhibited higher absolute values across all models (Fig. 6). RFECV also identified these features as key contributors to optimizing model performance (Fig. 7). These findings suggest that these coefficients significantly influence the

classification of target granular materials based on crater morphology. For example, five features (a , e , h , j , and o) showed a high CV score and well optimized the model trained on craters formed by a 10 mm drop (Fig. 7). Second-order polynomials primarily comprise the overall curvature of crater profiles, effectively modeling the typical convex shapes observed in craters. Fourth-order polynomials provide greater accuracy by accommodating more intricate curvature details (Fig. 2). These findings highlight that quantifying crater morphology by polynomial coefficients works as a reliable indicator of approximation of crater profiles for machine learning and that the properties and behavior of the given granular material significantly influence the shape of impact craters.

The confusion matrices for classification (Tables III–V) at conditions (a)–(c), except for the low-accuracy model at condition (d), highlight the challenges of distinguishing between specific granular materials. Notably, LHS-1 was frequently confused with MGS-1, while

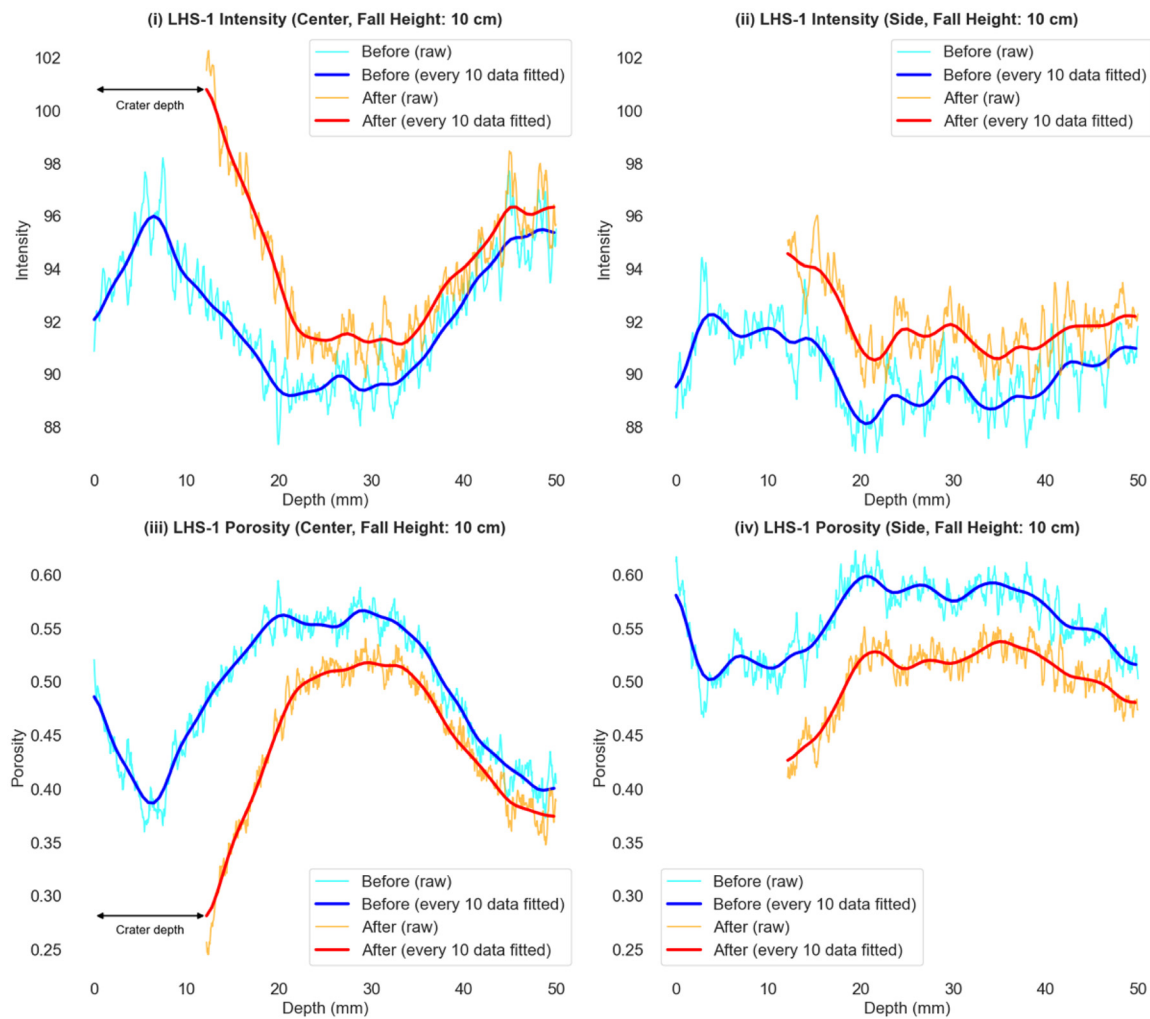


FIG. 13. Comparison of intensity profiles of LHS-1 between before and after impact (10 cm drop) in the studied area, (i) “center” and (ii) “side” (refer to Fig. 4). Comparison of hypothetical porosity profiles of LHS-1 between before and after impact (10 cm drop) in the studied area, (iii) “center” and (iv) “side” (refer to Fig. 4). The generated crater depth was 12.2 mm.

LMS-1 was confused with either MGS-1 or T-8. This misclassification can partly be attributed to the aspect ratio (AR) calculated by Eq. (2), which strongly influences crater morphology. For example, the aspect ratio of craters in LHS-1 media closely resembles that in MGS-1 media under most conditions, whereas LMS-1 exhibits an intermediate aspect ratio between MGS-1 and T-8 (Fig. 8). Furthermore, the average crater profiles derived from the coefficients (h , i , j , and k) using Eq. (5) show that the slope of crater sidewalls in LMS-1 media is similar to those of MGS-1 and T-8 media (Fig. 9). In contrast, T-8 was consistently well-classified and achieved higher F1 scores in our LR models under conditions (a)–(c). This reliable classification underscores the distinctiveness of the regolith simulants compared to terrestrial sand.

B. Slice analysis of CT datasets

Previous studies have established that crater depth and radius are influenced by target material properties such as porosity, cohesion,

and friction.^{1,11,18,32–35} To investigate these dependencies further and identify factors influencing target classification based on the crater formation, X-CT imaging was performed to analyze these material properties at both surface and subsurface levels. We obtained the CT intensity and hypothetical porosity profiles before and after impact in the studied granular media (Figs. 10–16).

When a fall height was 15.0 mm, the depths of generated craters were 5.2–7.5, 2.7–4.7, and 2.3–3.4 mm in the LHS-1, LMS-1, and T-8 granular media, respectively. For example, the porosity in “center” region at the given depth decreased by 0.06 and 0.01 on average (computed over the depths between the crater bottom and 30 mm) in the LHS-1 and LMS-1 granular media, respectively, due to the impact (Figs. 10 and 11). Notably, the LHS-1 granular media in “center” region showed the lowest porosity values at the depths near the crater bottom, around 0.20 lower than the highest value at a depth of ~ 36 mm (Fig. 10). Moreover, the porosity reduction in “side” region at

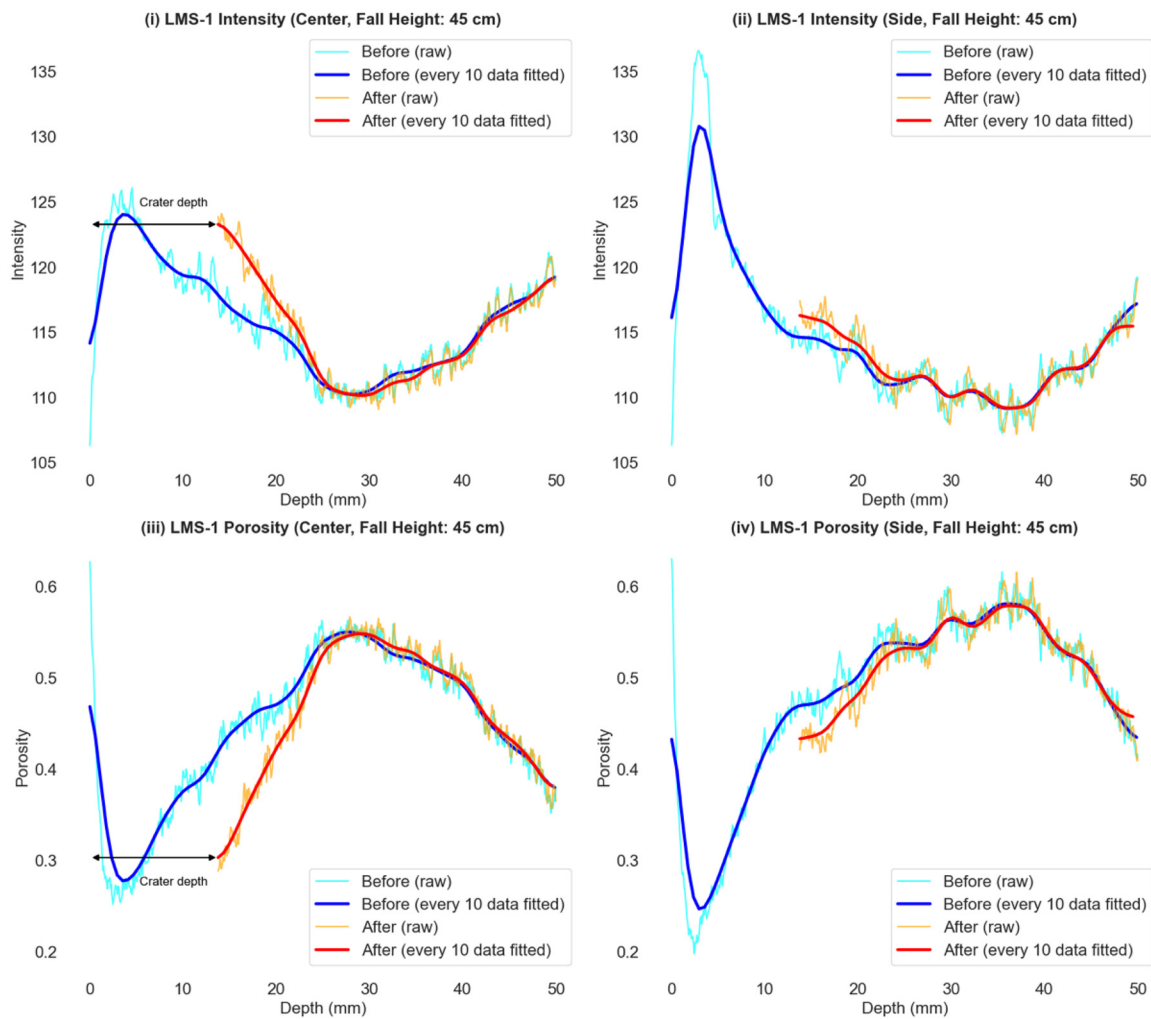


FIG. 14. Comparison of intensity profiles of LMS-1 between before and after impact (45 cm drop) in the studied area, (i) “center” and (ii) “side” (refer to Fig. 4). Comparison of hypothetical porosity profiles of LMS-1 between before and after impact (45 cm drop) in the studied area, (iii) “center” and (iv) “side” (refer to Fig. 4). The generated crater depth was 13.8 mm.

the given depth due to the impact is more significant, accounting for 0.07 and 0.02 on average (between the crater bottom and 30 mm depth) in the LHS-1 and LMS-1 granular media, respectively (Figs. 10 and 11). The porosity decrease due to the impact was more significant with higher drop height (i.e., higher impact velocity); for instance, the porosity at the given depth decreased by 0.04 and 0.06 on average (between the crater bottom and 30 mm depth) at “center” and “side” regions in the LMS-1 granular media, respectively, upon the projectile impact released from a height of 2.5 cm (Fig. 12). The porosity reduction due to the impact was even much greater with 0.10 and 0.04 on average (between the crater bottom and 30 mm depth) at “center” region in the LHS-1 and LMS-1 granular media, respectively, when a drop height was, respectively, 10 and 45 cm and the penetration depth was greater than the diameter of the spherical projectile (Figs. 13 and 14). Despite the spherical projectile being dropped onto LMS-1 media from a height of 4.5 times greater than that for LHS-1 media, the

resulting crater depths were 13.8 and 12.2 mm, respectively. These findings underscore LMS-1’s greater resistance to impact, attributed to its higher cohesion. In contrast, the T-8 media showed the opposite trends in porosity profiles, with that the porosity at the “center” region increased upon the impacts, although the decrease in porosity due to the impact was found in the “side” region (Figs. 15 and 16).

The 1.5 cm projectile drop into the surface of LHS-1 granular media, characterized by the highest mean porosity, formed the deepest crater in the experimental condition of $D < 2R$. This aligns with findings from earlier studies, which suggested a strong relationship between porosity and crater formation.^{11,18,32} Conversely, LMS-1, despite its slightly lower surface porosity than T-8, produced a smaller crater, likely due to its high cohesion (393 Pa) compared with the nearly noncohesive nature of T-8 (Table I). The shape factor of the studied particles (e.g., sphericity and elongation), potentially influencing the crater formation, is nearly identical among LHS-1, LMS-1, and

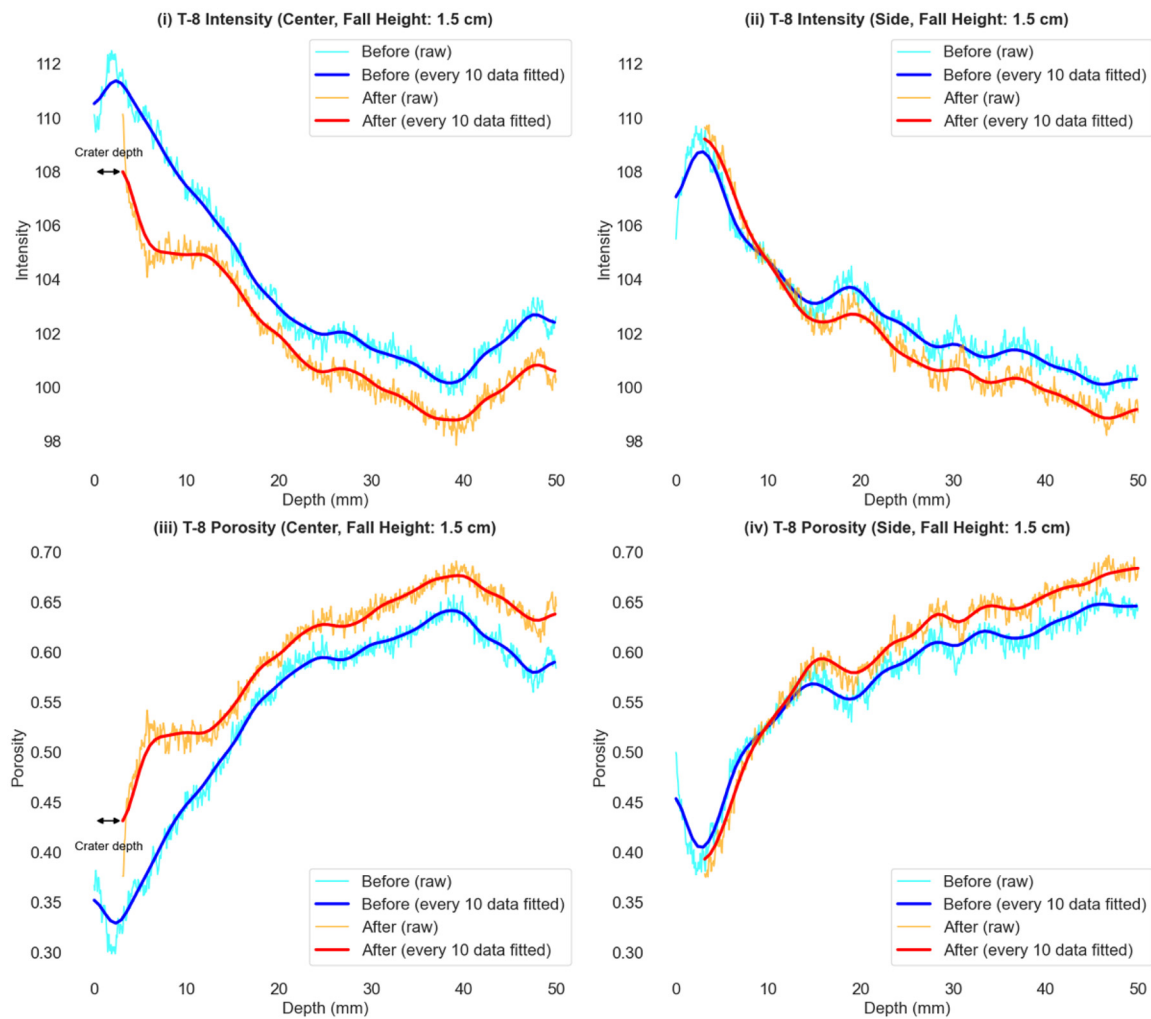


FIG. 15. Comparison of intensity profiles of T-8 between before and after impact (1.5 cm drop) in the studied area, (i) “center” and (ii) “side” (refer to Fig. 4). Comparison of hypothetical porosity profiles of T-8 between before and after impact (1.5 cm drop) in the studied area, (iii) “center” and (iv) “side” (refer to Fig. 4). The generated crater depth was 3.1 mm.

MGS-1 granular materials.²⁵ Our observation, thus, highlights the role of cohesion in influencing crater formation, with higher cohesion resulting in smaller craters, as previously reported.³³ To better investigate the observation, we further identified the particle distributions in the pre- and post-impacted granular media based on the intensity thresholds for the voids, enabling a visual comparison of particle distributions across different granular materials and between pre- and post-impact states (Figs. 17 and 18). For example, the cohesive properties of LMS-1 are visually evident in the particle distribution map (Fig. 17), where larger aggregates formed by particle adhesion are visible. The uppermost layer of LMS-1 media, where the particles accumulated, is also supposed to be caused by higher cohesion in addition to slight subsidence (Fig. 17). Moreover, the LHS-1 and LMS-1 granular media generally exhibited steeper porosity gradients than the T-8 media and strongly compacted (Figs. 10–16). These differences highlight the cohesive nature of regolith, significantly influencing the crater

morphology and subsurface profiles. Additionally, these cohesive characteristics of regolith can increase the force required to penetrate the subsurface, potentially complicating the drilling process on the lunar and Martian surfaces.

While taking into account the uncertainty of the intensity calibration, the post-impact porosity of the T-8 granular media increased (Figs. 15 and 16), deviating from the behavior observed in LHS-1 and LMS-1 representing the porosity decrease due to compression resulting from impact (Figs. 10–14). This may be due to ambient gas displacement (i.e., air) into the gap of noncohesive sand T-8 during impact, as observed by the increase in void spaces due to impact [Figs. 17(v) and 17(vi)]. This outcome also contrasts with previous studies employing discrete element method computations of the impact of steel projectiles on a cohesionless granular medium, which demonstrated the formation of a denser region beneath the projectile.³⁶ This discrepancy could be attributed to differences in the kinetic energy of the projectiles

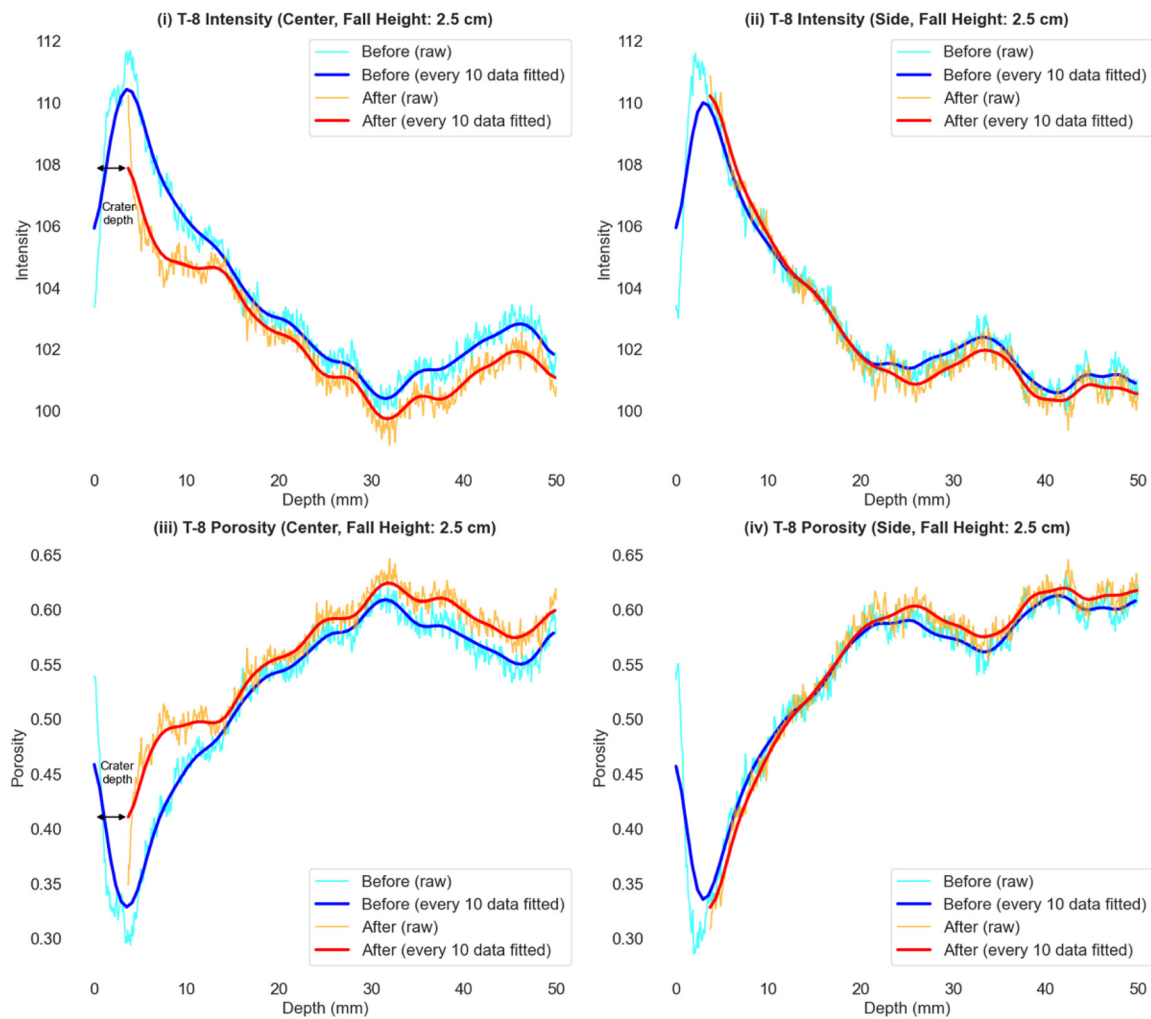


FIG. 16. Comparison of intensity profiles of T-8 between before and after impact (2.5 cm drop) in the studied area, (i) “center” and (ii) “side” (refer to Fig. 4). Comparison of hypothetical porosity profiles of T-8 between before and after impact (2.5 cm drop) in the studied area, (iii) “center” and (iv) “side” (refer to Fig. 4). The generated crater depth was 3.7 mm.

on the surfaces of granular media (0.3–0.5 mJ and 6.8–34 mJ in our X-CT study and the referenced study, respectively). Conversely, the porosity profiles of the LHS-1 and LMS-1 granular media showed a substantial reduction after impact (Figs. 10–14), with voids surrounding the spherical projectile being compressed when the penetration depth was greater than the diameter of the spherical projectile (Fig. 18). Notably, particle distribution maps from X-CT data well demonstrated that the LHS-1 granular materials in the immediate vicinity of crater bottom were subject to the substantial compaction, whereas it was not found in the other granular media (Figs. 17 and 18). These porosity and particle redistribution changes should relate to crater morphology, which might help classify the different granular materials from our LR machine learning model. This study bridges the gap between surface crater morphology and subsurface granular material behavior, which can also advance our understanding of lunar and Martian regolith mechanics. The findings of this study can establish a

baseline for understanding the behavior of fine-grained regoliths under the impact on celestial bodies, providing insights for the strategies and preliminary design of future exploration, drilling, and resource utilization on extraterrestrial terrains.

IV. CONCLUSION

This paper studied the granular behavior of lunar and Martian regolith simulants on low-velocity impact experiments through 3D surface scanning, machine learning-based classification, and X-CT imaging analysis. The classification by our LR machine learning could primarily differentiate studied granular materials, including regolith simulants (LHS-1, LMS-1, and MGS-1) and terrestrial sand (T-8), based on their crater profiles. Notably, the classification was moderately successful for the craters generated by different impact velocities, although the accuracy was up to 0.64. Features of the crater morphology, such as second- and fourth-order polynomial coefficients of the

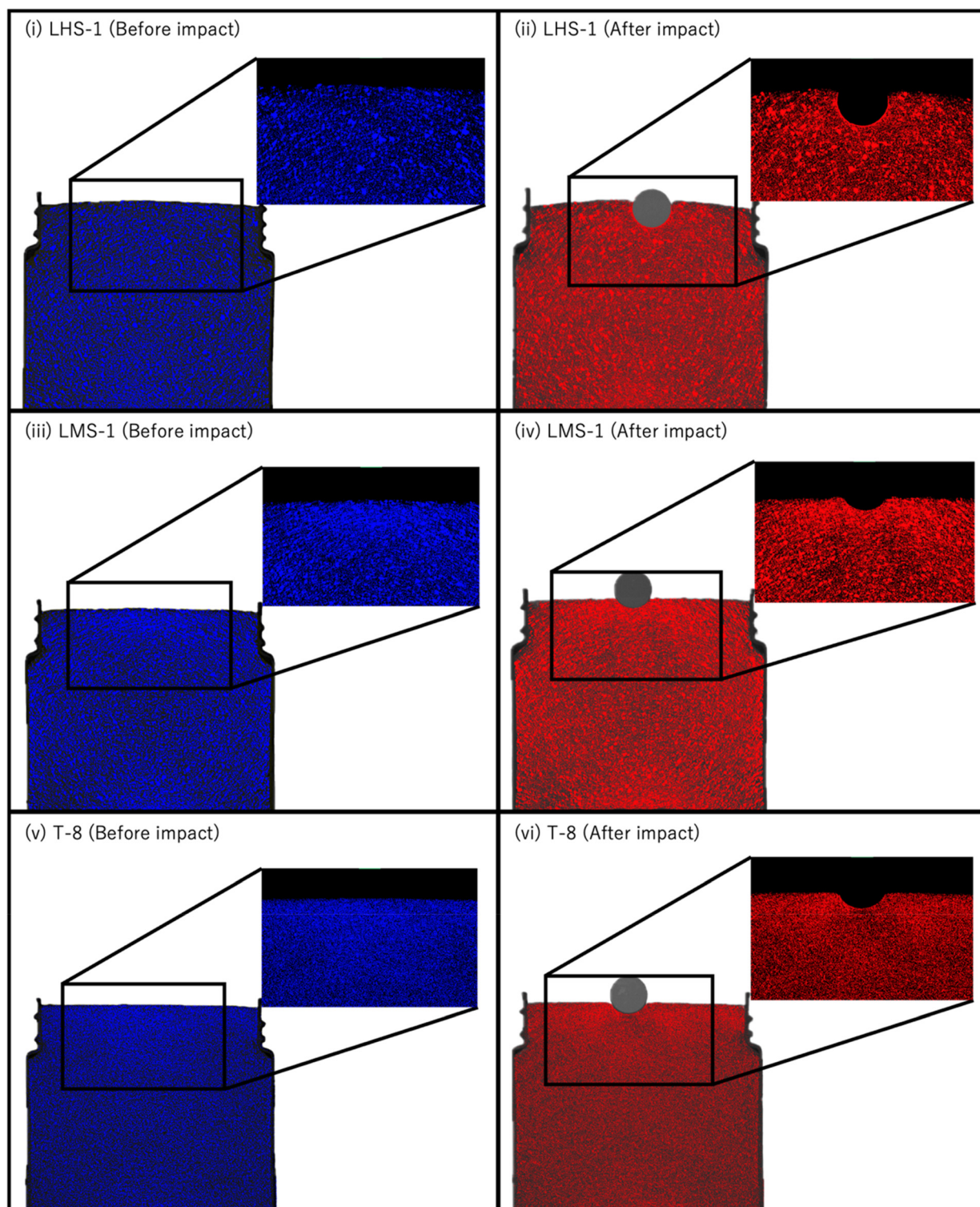


FIG. 17. Particle distribution of target material before and after a 1.5 cm drop. The assumed particles of LHS-1, LMS-1, and T-8 before impact were colored blue in (i), (iii), and (v), respectively. The assumed particles of LHS-1, LMS-1, and T-8 after impact were colored red in (ii), (iv), and (vi), respectively. The enlarged image focusing on the surface layer picked up only particles of each target material without a dropped sphere.

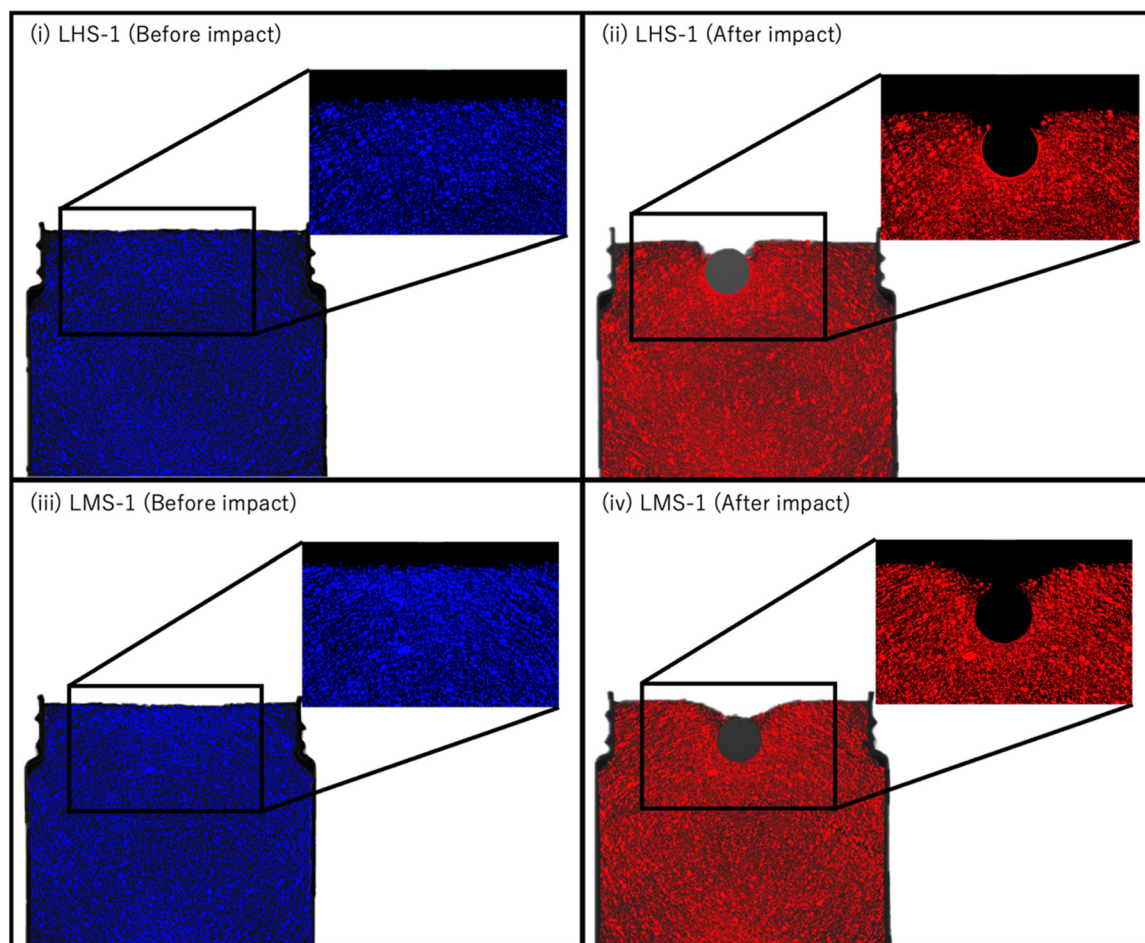


FIG. 18. Particle distribution of target material before and after the burial of a sphere. The assumed particles of LHS-1 and LMS-1 before impact were colored blue in (i) and (iii), respectively. The assumed particles of LHS-1 and LMS-1 after impact were colored red in (ii) and (iv), respectively. The enlarged image focusing on the surface layer picked up only particles of each target material without a dropped sphere.

given cross-sectional crater profiles, played pivotal roles in the classification, representing the deformation of the target layer and morphology of the given crater. Additionally, X-CT imaging revealed significant subsurface changes in porosity and particle distribution caused by impacts. For example, the porosity in the region beneath the crater at the given depth decreased by 0.20 on average in the LHS-1 granular media, even with the lowest velocity impact, which was greatest in the studied granular materials at the given impact velocity. This significant reduction in porosity agreed with a substantial rearrangement of particles evidenced by X-CT imaging. The LMS-1 granular media exhibited intermediate granular behavior between LHS-1 and T-8 granular media, reflecting its lower mean porosity in the shallower layer and higher resistance to impact due to the highest cohesion in the studied granular materials. The high cohesion mitigated surface deformation and influenced subsurface porosity changes, as evident in the gradual porosity decline at greater depths. The cohesive properties of LMS-1 also resulted in distinctive dappled patterns indicative of particle adhesion, distinguishing it from the noncohesive T-8 and the highly porous LHS-1. Our findings highlight the unique granular

properties of regolith simulants, distinguished them from terrestrial sand and revealing critical factors that influence their response to impact.

ACKNOWLEDGMENTS

The authors are grateful to S.-A. Cheng for his assistance throughout our RXCT measurements. This work was supported by JSPS KAKENHI Grant-in-Aid for Early-Career Scientists JP23K13698.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Takuma Ishii: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal);

Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Arata Kioka**: Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jyh-Jaan Steven Huang**: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Takeshi Tsuji**: Investigation (supporting); Writing – review & editing (supporting). **Yasuhiro Yamada**: Supervision (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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